Transported Romberg Extrapolation for the Arcsine Schröder Linearizer

Polynomial Composition Lifts and a Flat Infinite-Depth Limit

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Abstract

For an odd integer $m \geq 3$, let

$$S_m(z) = \sin(m \arcsin z), \qquad A(z) = \arcsin z,$$

so that S_m is an odd polynomial and $A(S_m(z)) = mA(z)$ as an analytic germ at 0. Starting from $P_1^{(m)}(z) = z$, consider the finite-polynomial recursion

$$P_{j+1}^{(m)} = P_j^{(m)} + \frac{mP_j^{(m)} - P_j^{(m)} \circ S_m}{m^{2j+1} - m}.$$

After the change of variable $z = \sin u$, this recursion is exactly a Richardson–Romberg polynomial extrapolation. It extrapolates $T_u(h) = \sin(hu)/h$ to $h = 0$ from the geometric nodes $h = 1, m, \ldots, m^{j-1}$. Consequently

$$P_j^{(m)}(\sin u) = \sum_{i=0}^{j-1} w_{j,i}^{(m)} \frac{\sin(m^i u)}{m^i},$$

with the standard Lagrange weights at zero in the variable h^2. The familiar jet-recovery and leading-defect formulas are therefore classical extrapolation consequences; in particular

$$\arcsin z - P_j^{(m)}(z) = \frac{m^{j(j-1)}}{(2j+1)!} z^{2j+1} + O(z^{2j+3}).$$

The new focus is the infinite-depth limit. The weights converge to

$$w_{\infty,i}^{(m)} = \frac{(-1)^i q^{i(i+1)/2}}{(q;q)_i (q;q)_\infty}, \qquad q = m^{-2},$$

which decay super-geometrically. Hence the limiting trigonometric series

$$F_m(u) = \sum_{i \geq 0} w_{\infty,i}^{(m)} \frac{\sin(m^i u)}{m^i}$$

is C^∞ on $\mathbb{R}$, and the finite tableaux converge to it together with every derivative. All moments required by polynomial extrapolation survive the limit, so $u - F_m(u)$ is flat at $u = 0$. The series is Hadamard-lacunary. Its Fourier coefficients decay super-algebraically but not exponentially; combined with the functional-differential identity

$$F_m(mu) = m\big(F_m''(u) + F_m(u)\big),$$

this shows that F_m is nowhere real analytic. The transported function $G_m = F_m \circ \arcsin$ is C^∞ on $(-1, 1)$ and differs from arcsin by a nonzero flat function at the origin. The cubic case $m = 3$ gives the scale-three specialization.

1 Introduction

Schröder's functional equation

$$\Psi(f(z)) = \lambda\Psi(z)$$

is a classical framework for local linearization near a fixed point, going back to Schröder's nineteenth-century work [1]. Richardson and Gaunt's deferred approach to the limit [2], and the geometric-grid formulation associated with Romberg [3], provide an equally classical mechanism for cancelling successive powers in an asymptotic expansion; see also Joyce's survey [4] and the monograph of Brezinski and Redivo-Zaglia [5]. A constructive computational treatment of Schröder linearization was given by Iserles [7], who also related the analysis to convergence acceleration of functional iteration. For a systematic modern treatment of Richardson extrapolation, including geometric and near-geometric extrapolation sequences and Romberg-type constructions, see Sidi [8].

The present paper concerns the intersection of these two classical structures. For every odd integer $m \geq 3$, define

$$S_m(z) := \sin(m \arcsin z). \tag{1}$$

The multiple-angle formula makes S_m an odd polynomial of degree m, and near the origin

$$A(S_m(z)) = mA(z), \qquad A(z) := \arcsin z, \qquad A(0) = 0, \quad A'(0) = 1. \tag{2}$$

For $m = 3$ this is the cubic sine-multiple map

$$S_3(z) = 3z - 4z^3.$$

Starting from $P_1^{(m)}(z) = z$, consider the recursion

$$P_{j+1}^{(m)}(z) = P_j^{(m)}(z) + \frac{mP_j^{(m)}(z) - P_j^{(m)}(S_m(z))}{m^{2j+1} - m}. \tag{3}$$

After writing $z = \sin u$, the recursion is precisely the Romberg/Neville tableau obtained by extrapolating

$$T_u(h) := \frac{\sin(hu)}{h}$$

to $h = 0$ through the nodes $h = m^i$. Thus the finite-stage jet recovery and its leading error constant are not a new extrapolation principle: they are the standard polynomial extrapolation mechanism transported through the nonlinear coordinate $z = \sin u$.

This identification leaves a more distinctive question. The geometric nodes move away from zero, rather than toward it, and the resulting Lagrange weights have a super-geometric limiting profile. Does the transported finite-polynomial tableau converge as its depth tends to infinity, and if it does, what is the limit? We prove that the tableaux converge in C^∞ in the linearizing variable to a periodic smooth function whose Taylor series at the origin is exactly u, but which is nowhere real analytic. The limit is a Hadamard-lacunary trigonometric series, placing it in a classical family studied in harmonic analysis; see Zygmund [9], Kahane [10], and Katznelson [11]. The distinctive feature here is that the full Taylor jet is prescribed exactly by the extrapolation moments. After transport back to z, the limiting polynomial sequence differs from $\arcsin z$ by a nonzero flat function.

Accordingly, the contribution claimed here is not Richardson/Romberg extrapolation itself, nor constructive approximation of Schröder linearizers, nor the multiple-angle identity. It is the exact transport of the classical geometric extrapolation tableau through the arcsine Schröder coordinate into a hierarchy of forward polynomial compositions, together with the explicit analysis of the resulting infinite-depth limit. In particular, the limiting object is smooth and has the full Taylor jet of the linearizer at the fixed point, yet differs from it by a nonzero flat function.

2 The transported Romberg tableau

Put

$$x_i := m^{2i}, \qquad i = 0, 1, \ldots,$$

and define the Lagrange weights for evaluation at $x = 0$ on the first j nodes by

$$w_{j,i}^{(m)} := \prod_{\substack{0 \le k < j \\ k \ne i}} \frac{x_k}{x_k - x_i}, \qquad 0 \le i < j. \tag{4}$$

For fixed u, set

$$\mathcal{T}_u(x) := \begin{cases} \dfrac{\sin(u\sqrt{x})}{\sqrt{x}}, & x \ne 0, \\ u, & x = 0. \end{cases} \tag{5}$$

Its power series

$$\mathcal{T}_u(x) = \sum_{r=0}^{\infty} \frac{(-1)^r u^{2r+1}}{(2r+1)!} x^r$$

shows that it is entire in x.

Theorem 2.1 (Transported Romberg representation)**.** *For every odd $m \ge 3$ and every $j \ge 1$,*

$$P_j^{(m)}(\sin u) = \sum_{i=0}^{j-1} w_{j,i}^{(m)} \frac{\sin(m^i u)}{m^i} = \sum_{i=0}^{j-1} w_{j,i}^{(m)}\, \mathcal{T}_u(x_i). \tag{6}$$

Equivalently, $P_j^{(m)}(\sin u)$ is the value at $x = 0$ of the degree-$(j-1)$ polynomial interpolating $\mathcal{T}_u(x)$ at $x = 1, m^2, \ldots, m^{2(j-1)}$. In the z variable,

$$P_j^{(m)}(z) = \sum_{i=0}^{j-1} w_{j,i}^{(m)}\, m^{-i} S_m^{\circ i}(z). \tag{7}$$

Proof. The interpolation formula gives the second expression in (6). Since $x_i = m^{2i}$, $\mathcal{T}_u(x_i) = \sin(m^i u)/m^i$.

It remains to identify the interpolation tableau with (3). Writing $\widehat{P}_j^{(m)}(u) := P_j^{(m)}(\sin u)$ and using $S_m(\sin u) = \sin(mu)$, the lift becomes

$$\widehat{P}_{j+1}^{(m)}(u) = \frac{m^{2j+1}\widehat{P}_j^{(m)}(u) - \widehat{P}_j^{(m)}(mu)}{m^{2j+1} - m} = \frac{m^{2j}\widehat{P}_j^{(m)}(u) - m^{-1}\widehat{P}_j^{(m)}(mu)}{m^{2j} - 1}. \tag{8}$$

To identify this recursion with the interpolation tableau, note first the scaling identity

$$\mathcal{T}_u(m^2 x) = m^{-1}\mathcal{T}_{mu}(x). \tag{9}$$

Hence the degree-$(j-1)$ interpolant of $\mathcal{T}_u$ on the shifted node set $x_1, \ldots, x_j$, evaluated at 0, is $m^{-1}\widehat{P}_j^{(m)}(mu)$ whenever $\widehat{P}_j^{(m)}(u)$ is the corresponding interpolated value on $x_0, \ldots, x_{j-1}$. Neville's two-block formula at $x = 0$, with $x_0 = 1$ and $x_j = m^{2j}$, therefore gives

$$\frac{x_j\widehat{P}_j^{(m)}(u) - m^{-1}\widehat{P}_j^{(m)}(mu)}{x_j - x_0} = \frac{m^{2j}\widehat{P}_j^{(m)}(u) - m^{-1}\widehat{P}_j^{(m)}(mu)}{m^{2j} - 1},$$

which is exactly (8). Since $\widehat{P}_1^{(m)}(u) = \sin u = \mathcal{T}_u(1)$ for all real u, induction proves the Lagrange formula (6) for all real u.

Finally, the polynomial identity $S_m^{\circ i}(\sin u) = \sin(m^i u)$ holds near $u = 0$ and therefore identically as a polynomial in $z = \sin u$, which gives (7). □

Remark 2.2. The theorem makes the classical content explicit. The finite polynomial lift is not merely analogous to Richardson extrapolation: after the coordinate change $z = \sin u$, it is exactly the same polynomial extrapolation tableau. The nonlinear composition structure appears when that tableau is transported back to z.

3 Finite-depth consequences

The classical exactness of polynomial interpolation immediately recovers the finite-order results.

Proposition 3.1 (Moment identities)**.** *For* $0 \le r \le j-1$,

$$\sum_{i=0}^{j-1} w_{j,i}^{(m)} x_i^r = \begin{cases} 1, & r = 0, \\ 0, & 1 \le r \le j-1. \end{cases} \tag{10}$$

Moreover,

$$\sum_{i=0}^{j-1} w_{j,i}^{(m)} x_i^j = (-1)^{j-1} \prod_{i=0}^{j-1} x_i = (-1)^{j-1} m^{j(j-1)}. \tag{11}$$

Proof. The identities in (10) follow from exact Lagrange interpolation of $1, x, \dots, x^{j-1}$ at $x = 0$. For x^j, let I_{j-1} denote its interpolating polynomial. The difference $x^j - I_{j-1}(x)$ is monic and vanishes at all nodes x_i, so

$$x^j - I_{j-1}(x) = \prod_{i=0}^{j-1} (x - x_i).$$

Setting $x = 0$ gives (11). □

Theorem 3.2 (Jet recovery and exact leading defect)**.** *For every odd* $m \ge 3$ *and every* $j \ge 1$,

$$P_j^{(m)}(z) = \arcsin z + O(z^{2j+1}), \tag{12}$$

and more precisely

$$\arcsin z - P_j^{(m)}(z) = \frac{m^{j(j-1)}}{(2j+1)!} z^{2j+1} + O(z^{2j+3}). \tag{13}$$

Also

$$\deg P_j^{(m)} = m^{j-1}. \tag{14}$$

Proof. Insert the power series of $\mathcal{T}_u$ into (6). The moment identities annihilate the terms x^r for $1 \le r \le j-1$, while (11) gives

$$P_j^{(m)}(\sin u) = u - \frac{m^{j(j-1)}}{(2j+1)!} u^{2j+1} + O(u^{2j+3}).$$

Thus $u - P_j^{(m)}(\sin u)$ has the displayed leading coefficient. Since $u = \arcsin z = z + O(z^3)$, transport back to z gives (12)–(13).

For the degree, (7) has a nonzero coefficient on $S_m^{\circ(j-1)}$, whose degree is m^{j-1}, while all earlier iterates have smaller degree. □

For the cubic case $m = 3$, the first residual coefficients are therefore

$$\frac{3^2}{5!} = \frac{3}{40}, \qquad \frac{3^6}{7!} = \frac{81}{560}, \qquad \frac{3^{12}}{9!} = \frac{6561}{4480},$$

for P_2, P_3, P_4, respectively. The first explicit lifts are

$$P_2(z) = z + \frac{1}{6}z^3, \tag{15}$$

$$P_3(z) = z + \frac{1}{6}z^3 + \frac{3}{40}z^5 - \frac{1}{10}z^7 + \frac{2}{45}z^9. \tag{16}$$

The degree-27 fourth lift begins

$$P_4(z) = z + \frac{1}{6}z^3 + \frac{3}{40}z^5 + \frac{5}{112}z^7 - \frac{1807}{1260}z^9 + O(z^{11}), \tag{17}$$

where the $O(z^{11})$ notation suppresses the remaining terms of the finite polynomial.

4 Intertwining with a geometric scale tower

The same tableau can be read directly in the scale variable. Suppose

$$N_{k+1} = mN_k, \qquad s_k = S_m(s_{k+1}), \tag{18}$$

with the points lying in a local branch on which (2) applies. Then

$$N_k A(s_k) = N_{k+1} A(s_{k+1}), \tag{19}$$

so $N_k \arcsin s_k$ is exactly invariant along the tower.

For $j \geq 1$, define

$$F_j(k) := N_k P_j^{(m)}(s_k)$$

and the level-j Richardson operator

$$(\mathcal{R}_j F)(k) := \frac{m^{2j} F(k+1) - F(k)}{m^{2j} - 1}. \tag{20}$$

Theorem 4.1 (Scale/composition intertwining identity)**.** *Whenever* (18) *holds,*

$$(\mathcal{R}_j F_j)(k) = F_{j+1}(k+1). \tag{21}$$

Proof. Using $N_{k+1} = mN_k$ and $s_k = S_m(s_{k+1})$,

$$\begin{aligned}
(\mathcal{R}_j F_j)(k) &= \frac{m^{2j} N_{k+1} P_j^{(m)}(s_{k+1}) - N_k P_j^{(m)}(s_k)}{m^{2j} - 1} \\
&= N_k \frac{m^{2j+1} P_j^{(m)}(s_{k+1}) - P_j^{(m)}(S_m(s_{k+1}))}{m^{2j} - 1} \\
&= mN_k P_{j+1}^{(m)}(s_{k+1}) \\
&= F_{j+1}(k+1).
\end{aligned}$$

□

For $m = 3$ and $j = 1$, writing $G_k = N_k s_k$ gives

$$\frac{9G_{k+1} - G_k}{8} = N_{k+1}\left(s_{k+1} + \frac{s_{k+1}^3}{6}\right),$$

the familiar first scale-three correction. The terminology "intertwining" is used deliberately: the identity has an index shift and is not a conjugacy in the strict dynamical-systems sense.

5 Infinite-depth weights

We now pass from the classical finite tableau to the infinite-depth problem. Put

$$q := m^{-2} \in (0,1), \qquad (q;q)_n := \prod_{r=1}^{n}(1-q^r), \qquad (q;q)_\infty := \prod_{r=1}^{\infty}(1-q^r). \tag{22}$$

The infinite product is strictly positive.

Proposition 5.1 (Closed finite and limiting weights)**.** *The Lagrange weights* (4) *satisfy*

$$w_{j,i}^{(m)} = \frac{(-1)^i q^{i(i+1)/2}}{(q;q)_i\,(q;q)_{j-1-i}}, \qquad 0 \le i < j. \tag{23}$$

For every fixed i,

$$w_{j,i}^{(m)} \longrightarrow w_{\infty,i}^{(m)} := \frac{(-1)^i q^{i(i+1)/2}}{(q;q)_i\,(q;q)_\infty}. \tag{24}$$

Moreover, uniformly in $j > i$,

$$|w_{j,i}^{(m)}| \le \frac{q^{i(i+1)/2}}{(q;q)_\infty^2}. \tag{25}$$

Proof. Since $x_i = q^{-i}$,

$$w_{j,i}^{(m)} = \prod_{k\neq i} \frac{1}{1-q^{k-i}}.$$

For $k < i$, put $r = i-k$ and use $1-q^{-r} = -q^{-r}(1-q^r)$; for $k > i$, put $r = k-i$. The two products give (23). Letting $j \to \infty$ yields (24). Finally $(q;q)_n \ge (q;q)_\infty > 0$ gives (25). □

The exponent $i(i+1)/2$ shows that the limiting weights decay faster than any geometric sequence. This compensates for the increasing frequencies m^i.

6 The smooth flat limit

Define

$$F_m(u) := \sum_{i=0}^{\infty} w_{\infty,i}^{(m)} \frac{\sin(m^i u)}{m^i}. \tag{26}$$

Theorem 6.1 (C^∞ convergence of the infinite tableau)**.** *For each integer $r \ge 0$, the series obtained by differentiating* (26) *r times converges absolutely and uniformly on $\mathbb{R}$. Consequently $F_m \in C^\infty(\mathbb{R})$, and*

$$P_j^{(m)}(\sin u) \longrightarrow F_m(u) \tag{27}$$

with uniform convergence of every derivative on $\mathbb{R}$.

Proof. Extend $w_{j,i}^{(m)}$ by zero for $i \ge j$. For fixed i, $w_{j,i}^{(m)} \to w_{\infty,i}^{(m)}$. For $r = 0$,

$$\left| w_{j,i}^{(m)} \frac{\sin(m^i u)}{m^i} \right| \le C q^{i(i+1)/2} m^{-i}.$$

For $r \ge 1$, the rth derivative of $\sin(m^i u)/m^i$ has absolute value at most $m^{i(r-1)}$, and hence

$$\left| w_{j,i}^{(m)} \frac{d^r}{du^r} \frac{\sin(m^i u)}{m^i} \right| \le C q^{i(i+1)/2} m^{i(r-1)} = C m^{-i^2+(r-2)i},$$

where $C = (q;q)_\infty^{-2}$. The majorant is summable for every fixed r. The Weierstrass test and Tannery's theorem (equivalently, dominated convergence for these coefficient arrays) give uniform convergence of the series and of every differentiated series. □

The finite interpolation moments survive the limit.

Proposition 6.2 (Infinite moment identities)**.** *For every integer $r \geq 0$,*

$$\sum_{i=0}^{\infty} w_{\infty,i}^{(m)} m^{2ri} = \begin{cases} 1, & r = 0, \\ 0, & r \geq 1. \end{cases} \tag{28}$$

Proof. Fix r. For all $j \geq r+1$, Proposition 3.1 gives the corresponding finite identity. The bound

$$|w_{j,i}^{(m)}| m^{2ri} \leq C q^{i(i+1)/2} q^{-ri}$$

is summable in i. Tannery's theorem therefore allows $j \to \infty$ and gives (28). □

Remark 6.3 (Euler's q-series form of the moments). The same identities follow directly from the classical Euler q-binomial identity

$$\sum_{i=0}^{\infty} \frac{(-1)^i q^{i(i-1)/2} y^i}{(q;q)_i} = (y;q)_\infty, \qquad |q| < 1,$$

see, for example, Gasper and Rahman [6]. Indeed, using $m^{2r} = q^{-r}$ and (24),

$$\sum_{i=0}^{\infty} w_{\infty,i}^{(m)} m^{2ri} = \frac{(q^{1-r};q)_\infty}{(q;q)_\infty}.$$

For $r = 0$ this ratio is 1, while for every $r \geq 1$ the numerator contains the factor $1 - q^0$ and therefore vanishes. This gives an independent q-series check of Proposition 6.2.

Theorem 6.4 (Flat and nowhere-analytic infinite-depth limit)**.** *The function F_m satisfies*

$$F_m'(0) = 1, \qquad F_m^{(r)}(0) = 0 \quad (r \neq 1). \tag{29}$$

Hence

$$H_m(u) := u - F_m(u) \tag{30}$$

is flat at $u = 0$. Nevertheless H_m is nonzero in every neighborhood of the origin, and F_m is nowhere real analytic on $\mathbb{R}$.

Proof. Even derivatives vanish because F_m is odd. For $r \geq 0$,

$$F_m^{(2r+1)}(0) = (-1)^r \sum_{i=0}^{\infty} w_{\infty,i}^{(m)} m^{2ri}.$$

Proposition 6.2 gives $F_m'(0) = 1$ and zero for every higher odd derivative. Thus H_m is flat.

To prove that the flat function is nonzero, first note from (24) that for $i \geq 1$

$$w_{\infty,i-1}^{(m)} = -(m^{2i} - 1) w_{\infty,i}^{(m)}. \tag{31}$$

Using the C^2 convergence from Theorem 6.1, termwise differentiation and an index shift give

$$\begin{aligned} F_m(mu) &= m\sum_{i=1}^{\infty} w_{\infty,i-1}^{(m)} \frac{\sin(m^i u)}{m^i} \\ &= -m\sum_{i=1}^{\infty} (m^{2i}-1) w_{\infty,i}^{(m)} \frac{\sin(m^i u)}{m^i} \\ &= m\big(F_m''(u) + F_m(u)\big). \end{aligned}$$

Thus

$$F_m(mu) = m\big(F_m''(u) + F_m(u)\big). \tag{32}$$

Suppose H_m vanished on some neighborhood of 0. Then $F_m(u) = u$ there, so $F_m''(u) = 0$ there. Equation (32) would then imply $F_m(v) = v$ on an interval m times larger. Iterating this argument gives $F_m(u) = u$ for every real u. But F_m is 2π-periodic, since every term in (26) is, whereas u is not. This contradiction proves that H_m is nonzero in every neighborhood of 0.

It remains to prove nowhere real analyticity. Write the nonzero Fourier coefficient of F_m at frequency $n_i = m^i$ as $\widehat{F}_m(n_i)$. From (24),

$$|\widehat{F}_m(m^i)| = \frac{|w_{\infty,i}^{(m)}|}{2m^i} = \frac{m^{-i(i+2)}}{2(q;q)_i(q;q)_\infty}.$$

Since $(q;q)_i \to (q;q)_\infty > 0$,

$$\log|\widehat{F}_m(m^i)| = -(i^2+2i)\log m + O(1) = -\frac{(\log n_i)^2}{\log m} - 2\log n_i + O(1). \tag{33}$$

Thus the Fourier coefficients decay faster than every algebraic power along the lacunary support, but they do not satisfy an exponential bound $O(e^{-c|n|})$ for any $c > 0$. A 2π-periodic real-analytic function on $\mathbb{R}$ has exponentially decaying Fourier coefficients; see, for example, Katznelson [11, Chapter I]. Hence F_m is not real analytic on all of $\mathbb{R}$.

Suppose now that F_m were real analytic on some nonempty open interval I. Equation (32) gives, for $v \in mI$,

$$F_m(v) = m\left(F_m''(v/m) + F_m(v/m)\right),$$

so F_m is real analytic on the dilated interval mI. Iteration gives analyticity on $m^k I$ for every $k \geq 1$. For sufficiently large k, the length of $m^k I$ exceeds 2π. Because F_m is 2π-periodic, analyticity on such an interval propagates by translation to all of $\mathbb{R}$, contradicting the preceding Fourier-coefficient argument. Therefore F_m is nowhere real analytic. □

Corollary 6.5 (Transported flat limit in the polynomial variable)**.** *Define*

$$G_m(z) := F_m(\arcsin z), \qquad -1 \leq z \leq 1. \tag{34}$$

Then G_m is continuous on $[-1,1]$, belongs to $C^\infty((-1,1))$, and

$$P_j^{(m)}(z) \longrightarrow G_m(z) \tag{35}$$

uniformly on $[-1,1]$ and in C^∞ on every compact subinterval of $(-1,1)$. Moreover,

$$\arcsin z - G_m(z) \tag{36}$$

is a nonzero flat function at $z = 0$. In particular, G_m is not real analytic at the origin despite having the same Taylor series there as $\arcsin z$.

Proof. Uniform convergence on $[-1,1]$ follows from Theorem 6.1 after putting $u = \arcsin z \in [-\pi/2, \pi/2]$. On compact subintervals of $(-1,1)$, composition with the smooth map arcsin preserves convergence of all derivatives. Indeed, with $H_m(u) = u - F_m(u)$ from (30),

$$\arcsin z - G_m(z) = H_m(\arcsin z).$$

Flatness follows because H_m is flat at 0 and $\arcsin z$ is analytic with $\arcsin z = z + O(z^3)$. If the flat difference vanished locally in z, then $F_m(u) = u$ locally after setting $z = \sin u$, contradicting Theorem 6.4. □

7 Cubic specialization and interpretation

For $m = 3$ one has $q = 1/9$ and

$$w_{\infty,i}^{(3)} = \frac{(-1)^i 9^{-i(i+1)/2}}{(1/9;1/9)_i (1/9;1/9)_\infty}. \tag{37}$$

The finite polynomials are therefore

$$P_j(z) = \sum_{i=0}^{j-1} w_{j,i}^{(3)} 3^{-i} S_3^{\circ i}(z), \qquad S_3(z) = 3z - 4z^3, \tag{38}$$

and converge on $[-1,1]$ to

$$G_3(z) = \sum_{i=0}^{\infty} w_{\infty,i}^{(3)} \frac{\sin(3^i \arcsin z)}{3^i}. \tag{39}$$

For scale, Table 1 records three evaluations at 50-digit precision (mpmath) of $H_3(u) = u - F_3(u)$. These values are only numerical illustrations; the nonvanishing and flatness statements above are proved independently. Every finite stage agrees with $\arcsin z$ to an increasing odd order, yet the

u	$H_3(u)$
0.1	$4.7381221300 \times 10^{-10}$
0.6	$2.2871746237 \times 10^{-3}$
1.5	$3.1540966034 \times 10^{-1}$

Table 1: Illustrative values of the nonzero flat discrepancy for $m = 3$.

infinite-depth limit is not $\arcsin z$. This resolves the convergence question left open by the finite-jet theorem: the sequence converges, but the limit contains a nonperturbative flat discrepancy invisible to every Taylor coefficient at the fixed point.

This phenomenon clarifies the roles of the different structures. The finite-depth coefficient cancellation is classical polynomial extrapolation. The polynomial composition formula is its exact transport through the multiple-angle dynamics. The flat nonanalytic limit appears only after taking the extrapolation depth to infinity on the outward geometric node set $1, m^2, m^4, \ldots$.

8 Scope and concluding remarks

The finite-depth statements in this paper should be read within classical Richardson/Romberg theory. The weights (4), the moment cancellations (10), and the leading error constant (11) are

standard consequences of polynomial extrapolation. Their appearance in (3) is explained exactly by Theorem 2.1.

The additional structure comes from the multiple-angle transport. It produces finite polynomials in z from an extrapolation tableau in h^2, gives the iterate formula (7), and makes the same operation appear as the scale/composition intertwining identity (21). At infinite depth the geometric Lagrange weights stabilize super-geometrically, which yields a global C^∞ Hadamard-lacunary trigonometric limit. The limiting moment identities force the full Taylor jet of the identity function. The Fourier-coefficient asymptotic (33), together with the functional-differential relation (32), strengthens the local nonanalyticity statement to nowhere real analyticity. Thus the limiting discrepancy is genuinely beyond all algebraic orders at the origin.

No claim is made that Richardson extrapolation, Romberg-type geometric tableaux, constructive Schröder approximation, Schröder linearization, or the multiple-angle identity is new. The paper's claim is the exact structural transport that turns the classical geometric extrapolation tableau into a forward finite-polynomial composition hierarchy for S_m, together with the analysis of its infinite-depth limit: a global smooth, nowhere-real-analytic lacunary trigonometric limit whose transport back to the polynomial variable has the complete arcsine Taylor jet at the origin but differs from arcsin by a nonzero flat function.

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